A-A-51149
25 October 1985

COMMERCIAL ITEM DESCRIPTION

CHUCKS, INDEPENDENT JAW

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-C-45833C.

1. SCOPE

1.1 Scope. This description covers wrench operated, four-jaw, independent type chucks with solid reversible jaws or two-piece reversible jaws of the standard, medium duty, commercial type, which are used in a production environment.

1.2 Classification. Chucks shall conform to class I, medium duty, in accordance with ANSI B5.8. Chucks shall be of the following types with the sizes as listed in table I. The type and size chuck required and the spindle nose type and size on which the chuck is to be mounted shall be as specified (see 6.2).

Type I - Chuck with solid reversible jaws.

Type II - Chuck with two-piece reversible jaws.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this description to the extent specified herein. The documents should be obtained from the contracting activity or as directed by the contracting officer.

Federal Specification PPP-T-1150	- Tools and Tool Accessories for Power Driven Metal and Woodworking Machinery, Packaging and Packing of.
Federal Standard FED-STD-H28	- Screw Thread Standards for Federal Services.
Federal Standard FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.

FSC 3460

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TABLE I. Chuck sizes and spindle nose type and size.

Chuck size, (inches)	USA Standard spindle nose type and size for mounting the chuck																			
6	A1-5	A2-5	B1-5	B2-5	A1-6	A2-6	B1-6	B2-6	D1-3	D1-4	L00	L0								
8	A1-5	A2-5	B1-5	B2-5	A1-6	A2-6	B1-6	B2-6	D1-3	D1-4	D1-5	D1-6	L00	L0	L1					
10	A1-5	A2-5	B1-5	B2-5	A1-6	A2-6	B1-6	B2-6	A1-8	A2-8	B1-8	B2-8	D1-3	D1-4	D1-5	D1-6	D1-8	L00	L0	L1
12	A1-5	A2-5	B1-5	B2-5	A1-6	A2-6	B1-6	B2-6	A1-8	A2-8	B1-8	B2-8	D1-5	D1-6	D1-8	L0	L1	L2		
14	A1-5	A2-5	B1-5	B2-5	A1-6	A2-6	B1-6	B2-6	A1-8	A2-8	B1-8	B2-8	D1-5	D1-6	D1-8	L0	L1	L2		
15	A1-6	A2-6	B1-6	B2-6	A1-8	A2-8	B1-8	B2-8	A1-11	A2-11	B1-11	B2-11	D1-6	D1-8	L0	L1	L2			
16	A1-6	A2-6	B1-6	B2-6	A1-8	A2-8	B1-8	B2-8	A1-11	B1-11	B1-11	B2-11	D1-6	D1-8	D1-11	L0	L1	L2		
18	A1-6	A2-6	B1-6	B2-6	A1-8	A2-8	B1-8	B2-8	A1-11	A2-11	B1-11	B2-11	D1-6	D1-8	D1-11	L0	L1	L2	L3	
20	A1-6	A2-6	B1-6	B2-6	A1-8	A2-8	B1-8	B2-8	A1-11	A2-11	B1-11	B2-11	D1-6	D1-8	D1-11	L1	L2	L3		
21	D1-8	D1-11	L2	L3																
22	A1-8	A2-8	B1-8	B2-8	A1-11	A2-11	B1-11	B2-11	D1-8	D1-11	L1	L2	L3							
24	A1-8	A2-8	B1-8	B2-8	A1-11	A2-11	B1-11	B2-11	A1-15	A2-15	B1-15	B2-15	D1-8	D1-11	L2	L3				
25	A1-8	A2-8	B1-8	B2-8	A1-11	A2-11	B1-11	B2-11	D1-8	D1-11	L2	L3								
28	A1-8	A2-8	B1-8	B2-8	A1-11	A2-11	B1-11	B2-11	A1-15	A2-15	B1-15	B2-15	D1-8	D1-11	L2	L3				
30	A1-11	A2-11	B1-11	B2-11	A1-15	A2-15	B1-15	B2-15	D1-8	D1-11	L3									
32	D1-8	D1-11																		
36	A1-11	A2-11	B1-11	B2-11	A1-20	A2-20	B1-20	B2-20	D1-11	L3										
42	A1-20	A2-20	B1-20	B2-20	D1-11	L3														

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ANSI B5.8	- Chucks, Chuck Jaws.
ANSI B5.9	- Spindle Noses.
ASTM D 3951	- Standard Practice for Commercial Packaging.

3. SALIENT CHARACTERISTICS

3.1 Design. The chucks shall be new and of the manufacturers standard commercial design in effect on the date of the solicitation. The chucks shall mount directly, or with the use of adapter mounting plates, onto lathes or other machines having spindle nose types A, B, D, or L and associated sizes in accordance with ANSI B5.9 and table I. Each jaw shall be adjusted separately with its own adjusting screw to facilitate holding workpieces. The chucks shall meet the requirements specified herein and shall conform to current industrial practice.

3.2 Materials. All parts and materials shall be of a quality necessary to produce a chuck to meet all requirements specified herein.

3.3 Chuck body. The chuck body shall be forged or cast steel alloy, semi-steel, ductile iron, or specially processed cast iron. The body shall be machined to accommodate the jaws, adjusting screws, thrust bearings, and mounting bolts. The chuck body dimensions, T-slots, bolt openings, and eye bolt holes shall be in accordance with ANSI B5.8. The face of the chuck shall have concentric ring grooves to aid in centering round workpieces.

3.4 Threads. All threaded parts shall conform to FED-STD-H28.

3.5 Chuck jaws. The type I chuck shall have four chuck jaws which shall be of one piece construction, individually adjustable, and shall be reversible for gripping externally and internally. All jaws shall be made of alloy steel and shall be hardened to not less than Rockwell "C" 58. Each jaw shall have two step bites and two end bites. The step bites and the end bites shall be serrated for all chuck sizes over 8 inches in diameter. Type II chucks shall have four two-piece (master jaw and top jaw) tongue and groove type jaws in accordance with ANSI B5.8. The jaws shall be made of alloy steel and the top jaw shall be hardened to not less than Rockwell "C" 58. The top jaw shall have two end bites and two step bites and shall be of the reversible type. The top jaw shall be attached to the master jaw with two socket head cap screws. The step bites and end bites of the top jaw shall be serrated. The bottom of the master jaws shall be provided with precision translating threads to accurately mate with the threads of the jaw adjusting screws.

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3.6 Adjusting screws. The chuck shall have four adjusting screws precision machined to mate with the threads of the corresponding chuck jaws. All adjusting screws shall be made of alloy steel and shall be hardened to not less than Rockwell "C" 58. The outer end of each adjusting screw shall be shaped to accept the applicable chuck wrench. All adjusting screws shall operate smoothly without binding or excessive play.

3.7 Chuck wrench. A chuck wrench, of the proper size and design, shall be furnished with each chuck. The wrench shall be of the T-handle design for manual adjustment of the chuck jaws. The chuck wrench shall be made of alloy steel and shall be hardened to not less than Rockwell "C" 40. When specified (see 6.2), additional chuck wrenches shall be furnished and fully described.

3.8 Thrust bearings. Each chuck jaw shall have thrust bearings which shall be made of alloy steel and shall be hardened to not less than Rockwell "C" 58.

3.9 Eye bolt. One or more eye bolts, for lifting the chuck, shall be furnished with each chuck 12 inches in diameter and larger. The eye bolt dimensions shall be in accordance with ANSI B5.8.

3.10 Mounting screws. The required number of socket head screws or bolts for mounting the chuck to the machine spindle or attaching the chuck to an adapter plate shall be furnished with the applicable chuck.

3.11 Runout tolerance. With the chuck mounted on a spindle having a spindle nose runout of not more than 0.0003 inch total indicator reading (TIR), the chuck runout shall be within 0.003 inch on the face and periphery during one complete rotation of the spindle. With a test bar having a diameter equal to the through hole in the chucks, mounted within the jaws, the runout shall be not more than 0.003 inch TIR in 360 degree rotation at 3 inches from the edge of the jaws.

3.12 Measurement system. All dimensions in this commercial item description are given in the English system of measurements. These dimensions can be converted to the Metric system of equivalents by the use of FED-STD-376.

3.13 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practical.

3.14 Workmanship. The workmanship of the chucks shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling inspection. The supplier shall randomly select the number of samples in table II that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination in 4.3, the dimensional inspection in 4.4 and the hardness test in 4.5. Any sample failing to meet all requirements specified herein shall be considered a defective item. If the number of defective items is equal to or less than the quantity specified under the "Acceptance, number defective" column of table II, the supplier shall replace the defective samples with items that meet all the requirements specified herein and the entire lot shall be considered acceptable for shipment. If the number of defective samples is equal to or greater than the quantity specified in the "Rejection, number defective" column of table II, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all chucks of the same type and size presented at one time for acceptance.

TABLE II. Sampling plan.

Quantity of items being purchased	Sample size	Acceptance, number defective	Rejection, number defective
2 - 8	2	0	1
9 - 15	3	0	1
16 - 25	5	0	1
26 - 50	8	0	1
51 - 90	13	0	1
91 - 150	20	1	2
151 - 280	32	1	2
281 - 500	50	2	3

4.3 Examination. Each chuck submitted for sample inspection shall be visually examined to determine conformance with all requirements of this description.

4.4 Dimensional inspection. Each of the sample chucks shall be inspected to assure that all dimensions are within the tolerances specified herein. Each sample chuck shall be inspected to determine conformance to the runout tolerance specified herein.

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4.5 Hardness test. After the sample inspection has been completed, a hardness test shall be performed on each of the test samples. If any of the test samples fail to meet the applicable hardness requirements specified herein, the total lot from which the samples were selected shall be rejected.

4.6 Packaging inspection. Packaging shall be inspected for conformance with section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or order, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2), preservation, packing, and marking shall be in accordance with Federal Specification PPP-T-1150, level as specified.

6. NOTES

6.1 Intended use. The chucks covered by this description shall mount directly or with the use of an adapter plate to an engine lathe or other machine to facilitate holding workpieces of irregular shapes where extreme accuracy is required.

6.2 Acquisition. Acquisition documents should specify the following:

a. Title, number, and date of this commercial item description.

b. Type, size, and quantity of chuck required and spindle nose type and size on which the chuck is to be mounted (see 1.2).

c. If additional chuck wrenches are required, the purchaser should specify the quantity required, manufacturer's name, catalog or other identifying numbers, and model number of chuck for which the wrench is intended (see 3.7).

d. Packaging in accordance with PPP-T-1150 if required, specify level required (see 5.1).

6.3 Cross reference data. The chuck sizes covered by this description are cross referenced to those previously covered in specification MIL-C-45833C.

MIL-C-45833C	A-A-51149
6 inch	(No change)
8 inch	(No change)
10 inch	(No change)
12 inch	(No change)
	14 inch (New size)

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MIL-C-45833C	A-A-51149
15 inch	(No change) 16 inch (New size)
18 inch	(No change) 20 inch (New size)
21 inch	(No change) 22 inch (New size)
24 inch	(No change) 25 inch (New size) 28 inch (New size) 30 inch (New size) 32 inch (New size) 36 inch (New size) 42 inch (New size)

MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Navy - AS
DLA - GS
Air Force - 84

User Activities

Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - IP

Project 3460-0105

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-51149	Chucks, Independent Jaw

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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